

UNIVERSITY OF CALIFORNIA IRVINE

PAUL MERAGE SCHOOL OF BUSINESS

A PROJECT REPORT ON

CUSTOMER SEGMENTATION ANALYSIS FOR SUN COUNTRY AIRLINES

FOUNDATION OF BUSINESS ANALYTICS

BANA 200

GROUP PROJECT

SUBMITTED TOWARDS THE PARTIAL FULFILLMENT OF THE REQUIREMENTS OF

MASTERS OF SCIENCE IN BUSINESS ANALYTICS

BY

NAMITH NAGARAJ

14569418

DELARAM DELPASSAND

22782472

MENGWEI ZHANG

65526131

YU-HSUAN LEE

33742810

ZIHAO LI

95863666

UNDER THE GUIDANCE OF

PROF. MINGDI XIN

INTRODUCTION

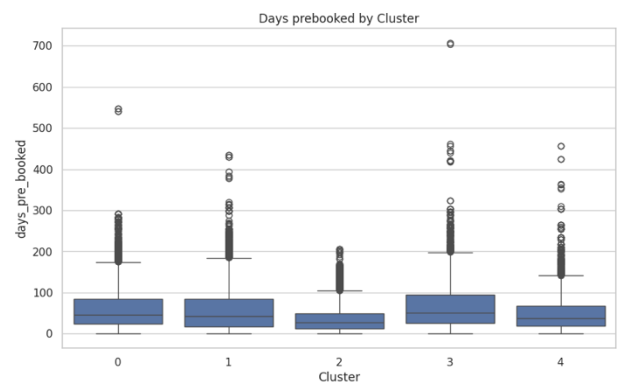
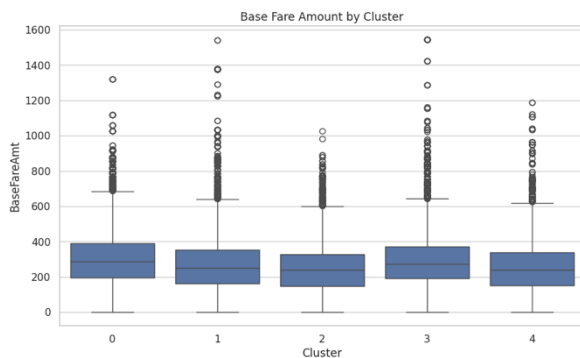
Sun Country Airlines seeks to gain a better understanding of its diverse customer base to enhance customer satisfaction, build loyalty, and improve overall business performance. This report provides a data-driven analysis, starting with descriptive statistics of key variables such as continuous data (e.g., Base Fare, Days Pre-booked) and categorical data (e.g., Ufly Membership, Booking Channel). These metrics are used to identify five distinct customer segments. Each segment's behavioural patterns are analysed to uncover their travel preferences and customer profiles. From these insights, we offer tailored recommendations aimed at improving customer experience and marketing strategies, leading to sustainable growth and profitability.

Based on your analysis, how many different customer segments (i.e., clusters of similar customers) are there? What are the key characteristics of each segment? Can you come up with descriptive titles for the segments?

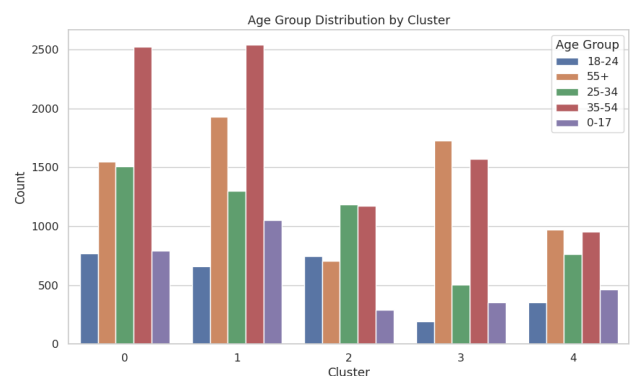
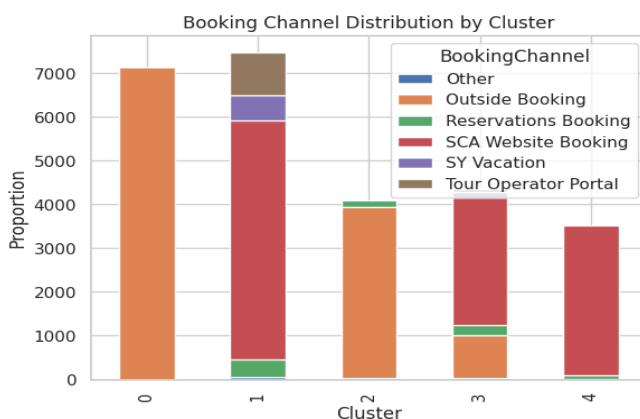
Descriptive Statistics

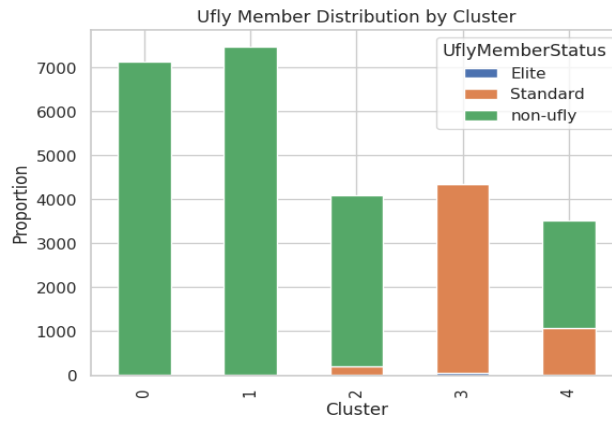
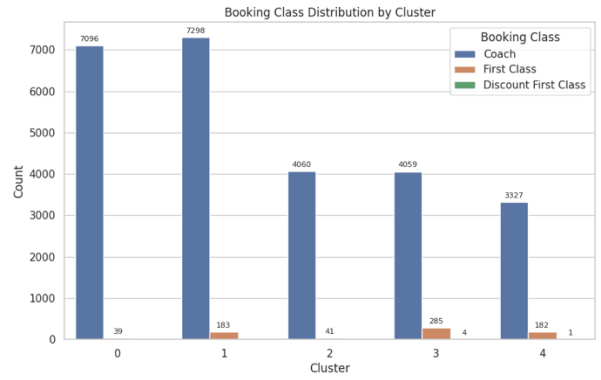
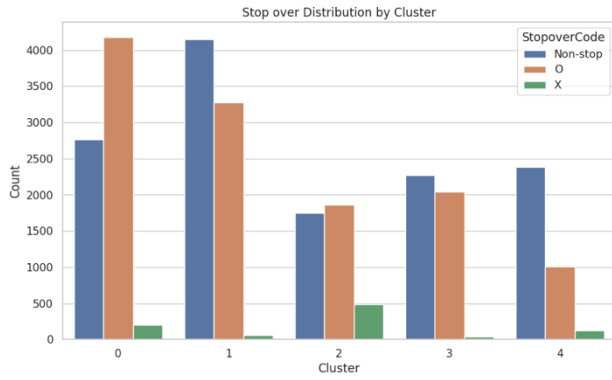
This section includes all the visualizations and descriptive statistics, which provide a clear foundation before diving into the five customer segments.

1. Continuous Variables

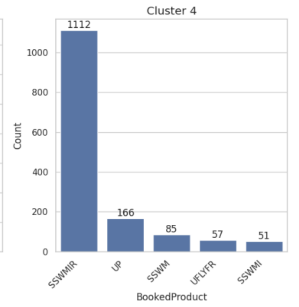
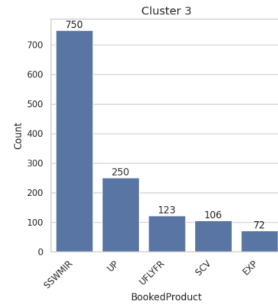
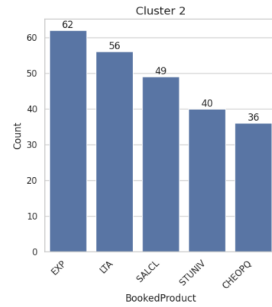
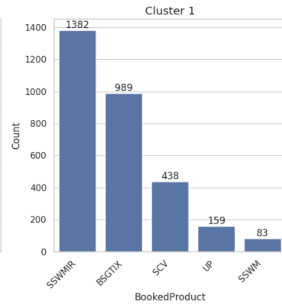
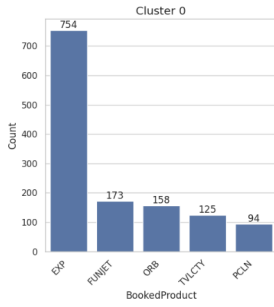


2. Categorical Variables

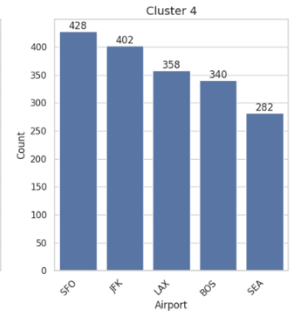
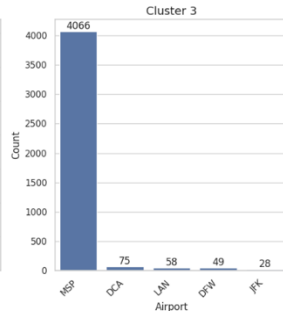
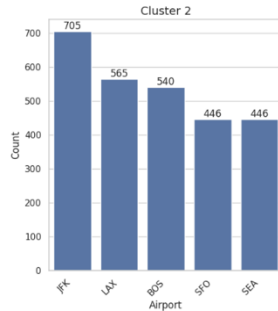
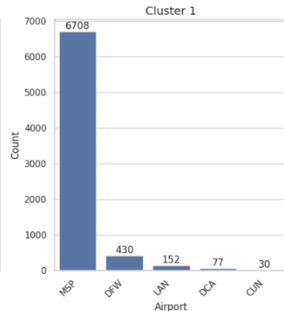
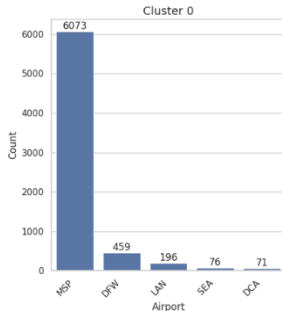




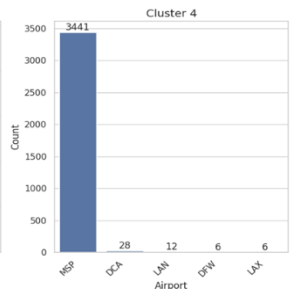
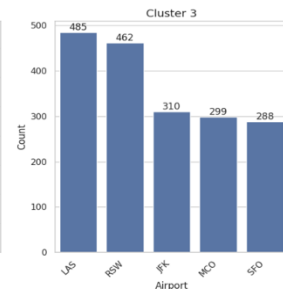
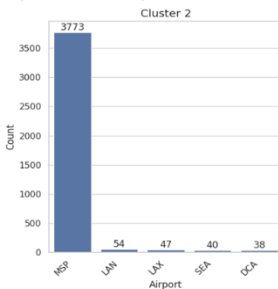
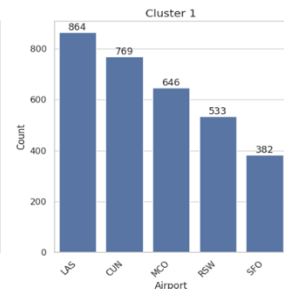
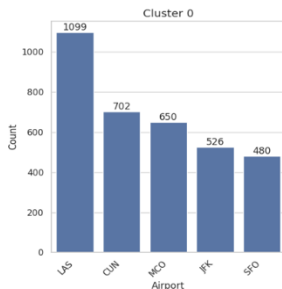
Top 5 BookedProduct Categories for Each Cluster



Top 5 Origin Airports for Each Cluster



Top 5 Destination Airports for Each Cluster

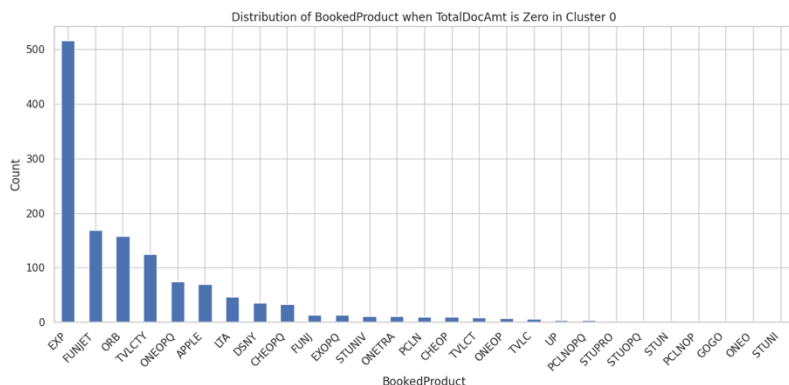


Five Customer Segments

Cluster 0: High-Spending Corporate Travelers with No Loyalty

Profile: Cluster 0 consists primarily of corporate travelers who pay higher base fares. They heavily rely on third-party booking channels and have minimal engagement with the airline's loyalty program. Many of their bookings have a total price of 0, likely due to corporate sponsorship, indicating they are not personally paying for their tickets.

- **Base Fare Distribution:** Referring to the Base Fare Amount by Cluster boxplot, Cluster 0 shows a higher average base fare, suggesting that these travelers are less concerned with cost due to company-covered expenses.
- **Booking Channel:** According to the Booking Channel Distribution, all travelers in this cluster rely heavily on Outside Booking platforms, likely due to corporate travel policies that utilize third-party systems for business trips.
- **Membership Status:** As seen in the Ufly Member Distribution, nearly all travelers in this group are **non-members**, further supporting the assumption that they are corporate travelers who do not personally handle or benefit from loyalty programs.
- **TotalDocAmt and BookedProduct:** The Distribution of BookedProduct when TotalDocAmt is 0 shows that the most common products linked to zero-priced bookings are EXP and FUNJET, which are likely corporate-sponsored travel products.



Cluster 1: Direct Channel Users with First Class Preferences but No Loyalty

Profile: This group primarily books directly through the airline's website and includes a small but notable portion of First-Class travelers. However, almost all of these First-Class travelers are not loyalty members, which presents an opportunity for engagement.

- **Base Fare Distribution:** The Base Fare Amount by Cluster boxplot shows that Cluster 1 travelers spend moderately on base fares, consistent with a mix of budget-conscious and mid-tier travelers.
- **Booking Channel:** As shown in the Booking Channel Distribution, the majority of Cluster 1 travelers use SCA Website Booking, indicating they prefer direct bookings through the airline's own platform.

- **Membership Status:** The Ufly Member Distribution indicates that the vast majority are non-members, even among First Class bookings in this cluster, highlighting a disconnect between premium spenders and loyalty engagement.

```
UflyMemberStatus counts for Cluster 1 First Class bookings:
UflyMemberStatus
non-ufly      182
Elite          1
Name: count, dtype: int64
```

Cluster 2: Younger Travelers with Short Booking Timeframes and High MSP Travel

Profile: This cluster represents younger, spontaneous travelers who tend to book trips closer to their departure date. They are mostly aged 25-34, with a noticeable group from the 18-24 range.

- **Base Fare Distribution:** Base Fare Amount by Cluster shows lower base fares, indicating they are price-sensitive, likely budget travelers.
- **Days Prebooked:** The Days Prebooked by Cluster data suggests they book relatively last-minute, often within 50 days of departure, which aligns with the spontaneous nature of these travelers.
- **Age Group Distribution:** Age Group by Cluster shows that this cluster consists mainly of younger travelers, particularly in the 25-34 age group, with a notable representation from the 18-24 group as well.
- **Destination Airports:** The Top 5 Destination Airports chart reveals that the vast majority of their travel is concentrated on flights to MSP, implying strong personal or professional ties to the region.

Cluster 3: Price-Conscious Loyal Travelers

Profile: Cluster 3 is made up of travelers who have strong loyalty to the airline but are price sensitive. These customers tend to book non-stop flights and may occasionally opt for premium services.

- **Membership Status:** The Ufly Member Distribution reveals that most travelers in this cluster are Standard members, showing strong engagement with the loyalty program, but they haven't yet reached elite status.
- **Booking Class:** The Booking Class Distribution indicates that most travelers in this cluster opt for Coach class, with only a few using First Class, signaling price-conscious behavior with occasional premium upgrades.
- **Stopover Code:** The Stopover Distribution shows a preference for Non-stop flights, which could imply a desire for convenience, particularly among loyal customers who may expect direct routes.

Cluster 4: Frequent MSP Travelers with Moderate Loyalty Engagement

Profile: Cluster 4 consists of frequent travelers who primarily fly to MSP from major US hubs. They book directly through the airline's website but have only moderate loyalty program engagement, remaining at the Standard membership level.

- **Base Fare Distribution:** The Base Fare Amount by Cluster boxplot shows that Cluster 4 has a moderate base fare, suggesting they are budget-conscious but travel frequently.
- **Booking Channel:** The Booking Channel Distribution highlights a strong preference for SCA Website Booking, showing trust in the airline's direct booking platform.
- **Membership Status:** The Ufly Member Distribution shows a large number of Standard members, indicating some loyalty engagement, but most travelers in this cluster have yet to advance to Elite status despite frequent travel.
- **Origin and Destination Airports:** According to the Top 5 Origin and Destination Airports, this group frequently flies from major US airports, especially SFO, JFK, and LAX, to MSP. This could indicate that these travelers are frequent commuters or have personal ties to the MSP area.

What advice would you offer the executives of Sun Country Airlines (Warnken and Vaughan), based on your segmentation analysis, so they can better achieve their business objectives?

Answer

- **For High-Spending Corporate Travelers (Cluster 0)**
 - **Boost Loyalty Program Engagement:**
 - **Work with Companies:** Partner with corporations to offer special deals like discounted bulk bookings or exclusive corporate memberships. This could make Sun Country the preferred choice for business trips.
 - **Tailor Offers:** Create loyalty programs specifically for corporate travellers, giving companies extra incentives to book through your airline.
 - **Create Corporate Travel Packages:** Offer Premium Perks i.e. Design travel packages with benefits such as lounge access and priority boarding. These perks can make your airline stand out for corporate bookings.
- **For Direct Bookers Who Prefer First Class but Aren't Loyal (Cluster 1)**
 - **Attract Premium Travelers:**
 - **Exclusive Benefits:** Offer special loyalty perks like bonus miles or First Class upgrades to encourage these travellers to join your loyalty program.
 - **Highlight Membership Benefits:** Emphasize the benefits of loyalty membership that align with their First Class preferences.

- **Encourage Direct Bookings:** Immediate Rewards: Provide instant rewards for travellers who book directly through your website, such as discounts or priority services, to boost both direct bookings and loyalty sign-ups.
- **For Younger Travelers Who Book Last-Minute and Often Fly to MSP (Cluster 2)**
 - **Promote Last-Minute Deals:**
 - **Engage Young Travelers:**
Youth-Friendly Loyalty Program: Create a loyalty program with immediate benefits like double miles or first-trip discounts to encourage younger travellers to join.
 - **MSP Travel Perks:**
 - **MSP Deals:** Since they frequently travel to MSP, offer special deals or rewards for MSP travel, such as discounted fares or priority boarding.
- **For Price-Conscious Loyal Travelers (Cluster 3)**
 - **Offer Upgrade Opportunities:**
 - **Try Higher Tiers:** Give loyal travellers chances to experience higher loyalty tiers or offer discounts on premium services to keep them engaged.
 - **Loyalty Discounts:**
 - **Exclusive Savings:** Provide special discounts on future bookings or bundle deals (like flights plus hotels) just for loyalty members to encourage continued loyalty.
- **For Frequent MSP Travelers with Moderate Loyalty (Cluster 4)**

Focus on MSP Benefits:

- **MSP-Specific Promotions:** Create promotions tailored to MSP travellers, such as discounted flights or exclusive perks like lounge access or fast-track boarding at MSP.
- **Speed Up Loyalty Progression** by providing Boost Status i.e., Offer ways to quickly advance in the loyalty program, like bonus points or tier matching, to encourage frequent MSP travellers to reach Elite status.

By implementing these strategies, Sun Country Airlines can better meet the needs of different traveller groups, increase loyalty, and drive more bookings.

Case Question:

1. How usable is the data you received from the client (Sun Country)? Given the state of the data, what steps would you take to make it usable and useful—for the initial assigned project and for Sun Country longer term?

Answer

The data received from Sun Country is already in a good state for analysis, thanks to prior pre-processing and the addition of dummy variables to handle missing values. This groundwork

simplifies further processing, visualization, and analysis, making it particularly suitable for clustering techniques like k-means.

For the initial assigned project, the key steps include verifying the accuracy of the dummy variables and ensuring data consistency. After that, descriptive analysis through visualizations and statistics will help uncover patterns in the data. Feature selection and scaling will prioritize the most relevant variables, especially important for clustering accuracy. Finally, clustering techniques will identify customer segments, and these results must be validated both from a data perspective and in alignment with business goals.

For Sun Country's long-term needs, data maintenance will be crucial, with processes established to regularly update the dataset and monitor data quality. Advanced analytics, such as predictive modeling and more sophisticated clustering methods, can further enhance insights over time. Additionally, developing interactive dashboards and regular reports will enable ongoing data monitoring and support decision-making. A continuous feedback loop with stakeholders will refine the data processes and ensure that insights remain aligned with evolving business objectives. These steps will ensure that the data remains a valuable resource for both current analysis and future strategic decisions at Sun Country.

2. Based on what you see in the data, how many ways can you view the information to paint a picture of different customer segments? Which ways offer Sun Country the most insights? How could the insights be used?

Answer

To analyze the customer segments, we used descriptive statistics and created various visualizations, including box plots, bar plots, pie charts, and more advanced filters to examine distribution patterns. For example, we focused on the distribution of the "TotalDocAmt" (the total price of the ticket, including the base fare) to see if a value of zero correlated with the use of discount codes. Additionally, we explored the characteristics of the top five departure cities ("true_origin") and destinations ("true_destination") to identify any unique features. Visualizations make it easier to spot differences and allow for deeper analysis of key variables, helping us uncover more actionable insights. These insights can then guide targeted marketing actions tailored to the characteristics of each segment.

3. How would you visualize and present the insights you found for the client? What will you do in order to show Warnken and Vaughan how the insights you derived connect to their business objectives?

Answer

To clearly present the insights to Warnken and Vaughan and show how they align with their business goals, we'll use a variety of visualization tools and strategies:

1. **Clustered Bar Charts:** We'll use these charts to compare different customer groups based on their spending, Ufly membership status, and group size. This will help us see how different types of customers behave and identify high-value customers who might need special marketing.
2. **Heatmaps and Pie Charts:** Heatmaps will show us where customers travel most frequently from major hubs like Minneapolis-St. Paul (MSP). This will help us understand customer preferences and adjust flight schedules or marketing efforts. Pie charts will reveal the percentage of customers in the Ufly Rewards program versus those who aren't, helping us find ways to boost membership.
3. **Promoting the Loyalty Program:** We'll use these insights to create targeted campaigns aimed at customers who aren't yet part of Ufly. By offering personalized deals and travel bundles, we can encourage more customers to join the loyalty program.
4. **Interactive Dashboards:** We'll set up an interactive dashboard that lets Warnken and Vaughan explore important data like customer engagement, booking channels, and average spending over time. This tool will make it easier for them to interact with the data and spot trends that could help improve customer experience and increase revenue.
5. **Customer Journey and Segmentation:** We'll provide visual tools like customer journey maps and segmentation dashboards to give a clear picture of customer preferences, booking habits, and online behaviour. This will help Warnken and Vaughan identify which customer segments offer the most growth potential.
6. **Connecting Insights to Business Goals:**
 - **Better Customer Insights:** We'll show how understanding different customer groups can help refine marketing strategies, such as targeting young, price-sensitive travellers or high-spending corporate clients with personalized offers.
 - **Increasing Loyalty Program Enrolment:** We'll explain how targeting non-Ufly members with exclusive benefits can boost enrolment in the rewards program.
 - **Improving Digital Experience:** We'll connect data insights to enhancements in the online booking experience, ensuring the website caters to tech-savvy, younger travellers.
 - **Developing Unique Products:** We'll illustrate how travel data can guide the creation of unique vacation packages, helping Sun Country stand out from bigger carriers with special, personalized offers.

By using these visual tools like bar charts, heatmaps, pie charts, interactive dashboards, and customer journey maps—we'll clearly show how the insights support Sun Country's goals. This approach will help Warnken and Vaughan make data-driven decisions to improve customer engagement, loyalty, and overall business performance.

4. Discuss how much power lies in simply understanding and exploring your data? Consider the downside of focusing too much on fancy models.

Answer

Power of Understanding and Exploring Data:

1. **Identifying Customer Segments:** Clustering revealed five distinct customer groups, helping Sun Country Airlines tailor marketing and loyalty programs to each group based on their booking habits and preferences.
2. **Actionable Insights Without Complexity:** Simple analysis showed that high-spending customers often book through the website but don't join the loyalty program. This insight allows for targeted loyalty perks without needing complex models.
3. **Immediate Business Impact:** Data exploration led to practical recommendations, such as offering deals to budget travellers and creating corporate packages, directly influencing pricing, loyalty programs, and marketing.

Downsides of Focusing on Fancy Models:

1. **Diminishing Returns on Complexity:** After a certain point, making a model more complicated may not significantly improve the quality of the insights and could even make them harder to interpret and use effectively. Basic clustering provided sufficient valuable information.
2. **Risk of Overfitting or Misaligned Insights:** Complex models can produce hard-to-use insights, while simple analysis delivered actionable recommendations easily.
3. **Increased Time and Resource Consumption:** Developing intricate models can delay decisions. Simple analysis provided timely, useful insights for quick business improvements.

In summary, the case study shows that straightforward data analysis can effectively reveal valuable insights and support business decisions without relying on complex models.